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Question 1: Social Capital

1.1 Merge Process Explanation:

To merge the Opportunity Atlas dataset with the Social Capital dataset, I first identified that the two datasets used different county coding systems. The Atlas dataset provides separate FIPS codes for state and county, while the Social Capital dataset provides a combined FIPS code. I constructed a unified FIPS identifier in the Atlas dataset using:

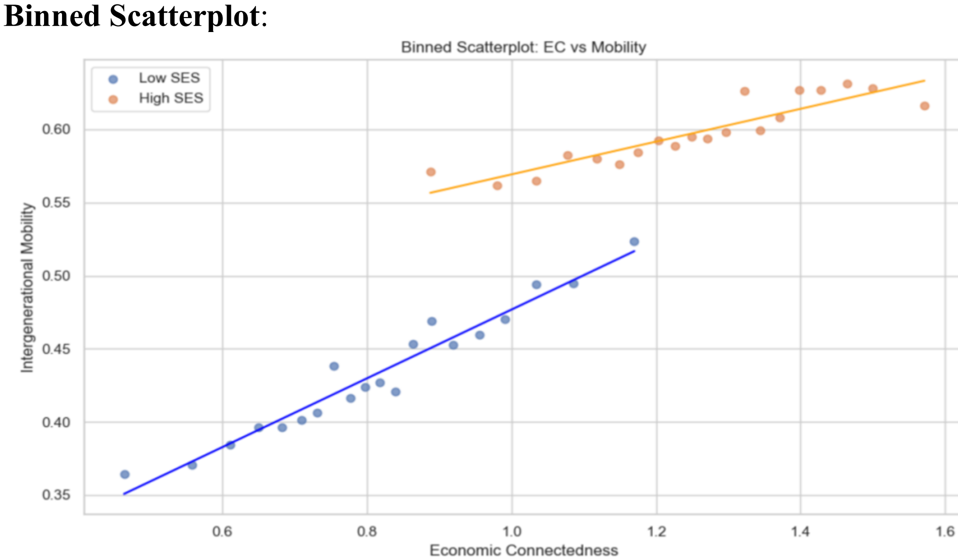
$$\text{fips} = \text{state } \# \cdot 1000 + \text{county } \#$$

I then renamed the Social Capital county code to fips and performed an inner merge on this key. The final merged dataset contains 3,141 rows, representing all U.S. counties and county equivalents. This merged dataset was saved as output/q1_merged_dataset.csv and is included in the ZIP submission.

1.1 Summary Statistics Table:

	mean	min	25%	50%	75%	max	n
kfr_pooled_pooled_p25	0.433798	-0.144350	0.386554	0.421304	0.461266	4.959184	3141
kfr_pooled_pooled_p75	0.599083	-0.156952	0.563935	0.590666	0.622236	4.791885	3141
ec_county	0.815030	0.294690	0.696742	0.807775	0.938373	1.359700	3070
ec_high_county	1.253287	0.700620	1.135420	1.258635	1.384657	1.715070	3070

1.2 Binned Scatterplot:



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Low SES slope: 0.23501271617821537
Low SES SE: 0.01271958234065491
High SES slope: 0.11207838863486977
High SES SE: 0.012420086975764936
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1.2 Interpretation of Binned Scatterplot:

The binned scatterplot displays a strong positive association between economic connectedness and intergenerational mobility for both low- and high- socioeconomic status (SES) groups. In counties where low-SES individuals maintain greater cross-class social ties with high-SES peers, children from disadvantaged backgrounds demonstrate substantially higher rates of upward mobility. This relationship is reflected in the steeper regression slope for the low-SES group (slope ≈ 0.235, with an SE ≈ 0.0127), indicating that increases in economic connectedness are strongly correlated with improved mobility outcomes. While high-SES individuals also display a positive slope (slope ≈ 0.112, SE ≈ 0.0124), the magnitude of the estimate is less than half as large, suggesting that economic connectedness matters more for those starting from lower socioeconomic positions. These findings overall provide compelling empirical evidence that cross-class social ties serve as a primary predictor of upward economic mobility.

1.3 Three Necessary Assumptions:

We are given the following statement by a local official: "We want to improve intergenerational mobility in our county. Schools in our county are segregated by income. Because economic connectedness is positively associated with intergenerational mobility, a policy that allocates students to schools so that each class has a mix of rich and poor students will improve intergenerational mobility in our county.". For this statement to be true, the following three assumptions are necessary:

1. Causal Effect of Economic Connectedness – We must assume that the positive association between economic connectedness and upward mobility reflects a causal relationship rather than just correlation. If there is a causal relationship between these two factors, then increasing cross-class ties causes upward mobility. However, if there is just a correlation between the two factors, then some outside third factor, such as county-level economic growth or quality of education, could be driving this positive connection.

2. Exposure Leading to Interaction – It must also be assumed that the proposed policy implementation of mixing rich and poor students in a classroom will generate meaningful social ties between classes. If students self-segregate in classrooms by forming friendships within their respective SES-groups, or the schools fail to encourage interaction between SES-groups, then this policy only changes classroom composition and will not change economic connectedness. As such, the policy would do nothing to benefit upward economic mobility.

3. External Validity (Scale and Setting Transferability) – Our data has only shown that economic connectedness at a county-wide level correlates to upward economic mobility. For the suggested policy to succeed, we must assume that this county-wide context also applies in a classroom setting without triggering unintended offsetting effects such as high-SES families enrolling in private schools instead of public education.

Question 2: Measurement Error and Attenuation Bias

2.1: We want to show that $\lim_{n\to\infty} \hat{\beta}^* = \beta \cdot \frac{Var(Income_i)}{Var(Income_i)+Var(u_i)}$

Starting with the OLS slope estimating formula: $\hat{\beta}^* = \frac{Cov(Income_i^*, TestScore_i)}{Var(Income_i^*)}$

We are given the true model: $TestScore_i = \alpha + \beta Income_i + \epsilon_i$

And mismeasured income: $Income_i^* = Income_i + u_i$

So, from the OLS slope estimator, we get that the numerator becomes the following:

$$Cov(Income_i^*, TestScore_i) = Cov(Income_i + u_i, \alpha + \beta Income_i + \epsilon_i)$$

We can then expand the covariance term by term to get the following:

$$\begin{aligned} &Cov(Income_i^*, TestScore_i) \\ &\qquad = Cov(Income_i, \alpha) + \beta Cov(Income_i, Income_i) + Cov(Income_i, \epsilon_i) \\ &\qquad + Cov(u_i, \alpha) + \beta Cov(u_i, Income_i) + Cov(u_i, \epsilon_i) \end{aligned}$$

We know that $Cov(u_i, Income_i) = 0$, $Cov(u_i, \epsilon_i) = 0$, $Cov(Income_i, \epsilon_i) = 0$, $Cov(Income_i, Income_i) = Var(Income_i)$, and any Covariance of constants = 0

As such all terms in our numerator drop out, leaving just the following:

$$Cov(Income_i^*, TestScore_i) = \beta Var(Income_i)$$

Now onto the denominator: $Var(Income_i^*) = Var(Income_i + u_i) = Var(Income_i) + Var(u_i)$

Combining the numerator and denominator gives us the following:

$$\hat{\beta}^* = \frac{Cov(Income_i^*, TestScore_i)}{Var(Income_i^*)} = \frac{\beta Var(Income_i)}{Var(Income_i) + Var(u_i)}$$

We can then take the limit as n → ∞ to get $\lim_{n\to\infty} \hat{\beta}^* = \beta \cdot \frac{Var(Income_i)}{Var(Income_i)+Var(u_i)}$, as desired.

This shows that measurement error biases the OLS slope towards zero as the denominator is always greater than the numerator.

2.2: We define the signal-to-noise ratio as $SNR = \frac{Var(Income_i)}{Var(Income_i)+Var(u_i)}$

This means that $\hat{\beta}^* = \beta \cdot SNS$. The SNR compares the real variation in true income (signal) against the variation in income coming from measurement error (noise). When there is a larger amount of noise in comparison to the signal, the OLS is less informed by the true income, and thus the slope flattens out. Essentially, a high SNR means there is little bias in the data, while a low SNR translates to more severe bias.

When $Var(u_i) \to 0$, the SNR becomes $SNR = \frac{Var(Income_i)}{Var(Income_i)+0} = 1$. Thus $\hat{\beta}^* = \beta \cdot 1 = \beta$. This is an example of perfect measurement with no attenuation bias, where the OLS is able to recover the true causal effect.

When $Var(u_i) \to \infty$, we get $SNR = \frac{Var(Income_i)}{Var(Income_i)+\infty} \to 0$, and thus $\hat{\beta}^* = \beta \cdot 0 = 0$. This is an extreme example of the measured income being almost entirely noise, and as such OLS has a slope of 0 and shows no relationship between income and test scores. This is the most extreme form of attenuation bias.

2.3: Results Table

	n	sigma_u	Expected E[beta*]	Avg Estimated beta*	Avg OLS SE
0	250	0	0.002000	0.001998	0.000635
1	250	5000	0.001600	0.001575	0.000569
2	250	15000	0.000615	0.000611	0.000357
3	1000	0	0.002000	0.002003	0.000317
4	1000	5000	0.001600	0.001586	0.000284
5	1000	15000	0.000615	0.000610	0.000178
6	5000	0	0.002000	0.001997	0.000141
7	5000	5000	0.001600	0.001601	0.000127
8	5000	15000	0.000615	0.000613	0.000080

How the estimated coefficient $\hat{\beta}^*$ changes as σ_u (sigma_u) increases:

- When $\sigma_u = 0$, there is no measurement error and our average estimated $\hat{\beta}^* \approx 0.002$.
- When $\sigma_u = 5000$, there is a moderate amount of measurement error which leads to moderate attenuation. As such, our average estimated $\hat{\beta}^*$ consistently drops to between 0.0015 and 0.0016.
- When $\sigma_u = 15000$, there is a large measurement error and simultaneously a severe attenuation. Our average estimated $\hat{\beta}^*$ drops to about 0.0006 in all cases.

From this, we can conclude that more measurement error results in a lower estimated slope and thus stronger attenuation bias.

How the average OLS standard error changes as n increases:

- When n = 250, our average OLS standard errors are the three highest in this regression simulation, at 0.000635, 0.000569, and 0.000357 for $\sigma_u = 0$, $\sigma_u = 5000$, and $\sigma_u = 15000$ respectively.
- When n = 1000, the average OLS standard errors shrink to 0.000317, 0.000284, and 0.000178 for $\sigma_u = 0$, $\sigma_u = 5000$, and $\sigma_u = 15000$ respectively.
- When n = 5000, the average OLS standard errors continue to shrink and become quite small, at 0.000141, 0.000127, and 0.000080 for $\sigma_u = 0$, $\sigma_u = 5000$, and $\sigma_u = 15000$ respectively.

This leads us to the conclusion that a larger sample size results in a smaller standard error and more precise estimate, as is statistically logical.

2.4: Including parental education affects both the value and interpretation of $\hat{\beta}$. Since standard prior economic studies have shown that parental education is typically correlated with true income, controlling for Edu_i removes the portion of income variation that is predictable from education. This reduces the signal in income while the measurement error (noise) remains unchanged. As a result, the signal-to-noise ratio falls, and attenuation bias becomes more severe. Therefore, the estimated coefficient $\hat{\beta}$ becomes even smaller than in the regression without parental education.

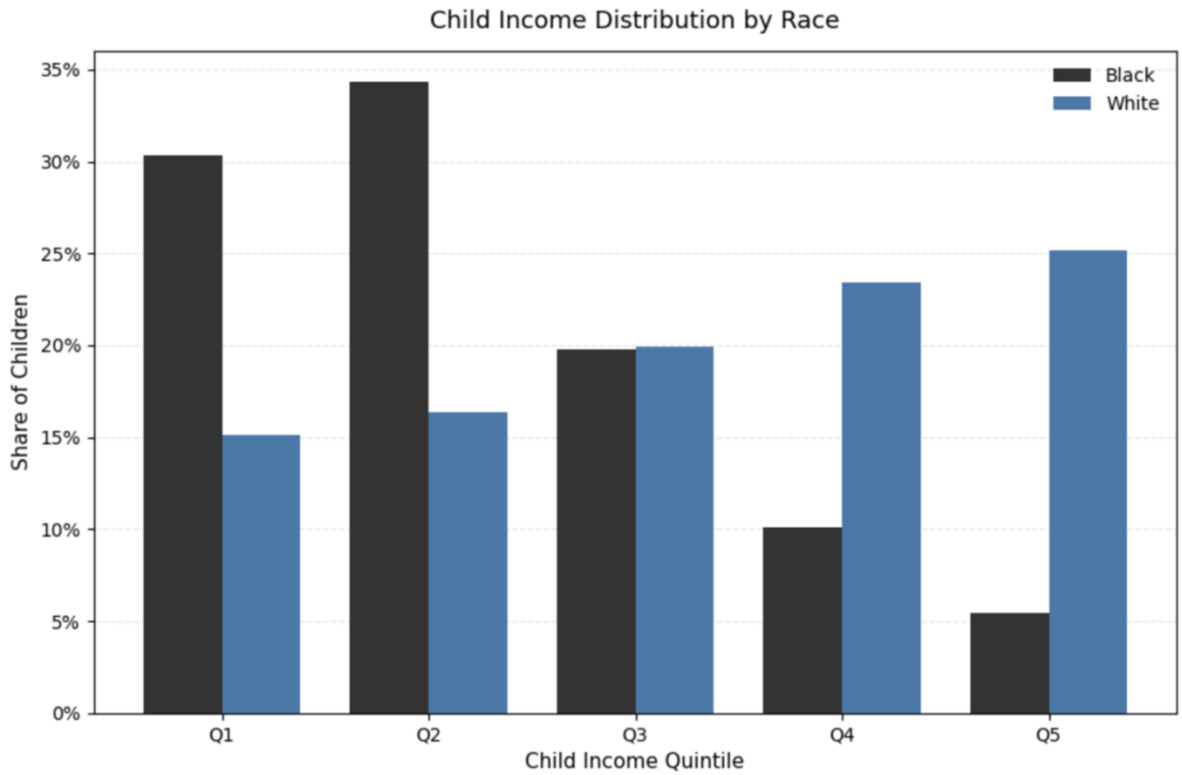
In terms of interpretation, $\hat{\beta}$ no longer reflects the total effect of income on test scores. Instead, it measures the partial effect of income holding parental education fixed. Since parental education explains part of the variation in test scores that income previously captured, the coefficient on income shrinks both because of attenuation bias and because the regression attributes some of income's explanatory power to education.

2.5: One example where survey-based measures are likely to suffer from severe measurement errors is a self-reported query on household income. Respondents may misremember their total income, not account for the income of all household members, round their income, confuse pre- and post- tax income, or intentionally deceive the survey due to privacy concerns. To circumvent the measurement errors this would cause, a researcher could use administrative tax records, census data, and/or employer-verified earnings as an external source of data. This would capture all income sources with far less bias. If such data is unavailable or unable to be accessed, the researcher could also use an instrumental variable, such as industry earnings changes or policy driven earnings changes. This shows variation in true income without any measurement error, eliminating attenuation bias entirely.

Many other examples of survey-based measures that are likely to suffer from measurement errors exist, including self-reported expenditures, health/disability status, or hours worked.

Question 3: Intergenerational Mobility

3.1: Bar Graph

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Child Income Distributions
==================================================

Black
Q1: 0.3032
Q2: 0.3432
Q3: 0.1978
Q4: 0.1012
Q5: 0.0545
Total: 1.0000

White
Q1: 0.1512
Q2: 0.1635
Q3: 0.1995
Q4: 0.2344
Q5: 0.2514
Total: 1.0000
```

Interpretation:

The unconditional income distribution differs substantially by race. Black children are more concentrated in the lower income quintiles and are substantially less represented in the top income quintile, while White children have greater representation in the upper income quintiles. These differences reflect both differences in the parental income distributions and differences in the race-specific intergenerational transition matrices.

3.2: To recover the income distribution of the parents' parents (the grandparents), we use the fact that the parent generation is assumed to have the same intergenerational transition matrix as the children. The forward mobility relationship is

$$\text{ParentDist} = T \times \text{GrandparentDist}$$

Since we observe ParentDist and know the transition matrix T, we can solve for the grandparents' income distribution by inverting the matrix (assuming that T is invertible):

$$\text{GrandparentDist} = T^{-1} \times \text{ParentDist}$$

Thus, for each race, the procedure is: (1) take the race-specific transition matrix, (2) compute its inverse, and (3) multiply the inverse by the observed parent income distribution. This yields the implied income distribution of the grandparents under the assumption of identical mobility patterns across generations.

3.3: In a universe with no income mobility, every child remains in the same income quintile as their parents. Therefore, the intergenerational transition matrix is the 5×5 identity matrix. Each diagonal entry equals 1, indicating that children always stay in their parents' quintile, and all off-diagonal entries are 0, indicating that movement between quintiles never occurs.

$$T_{no-mobility} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

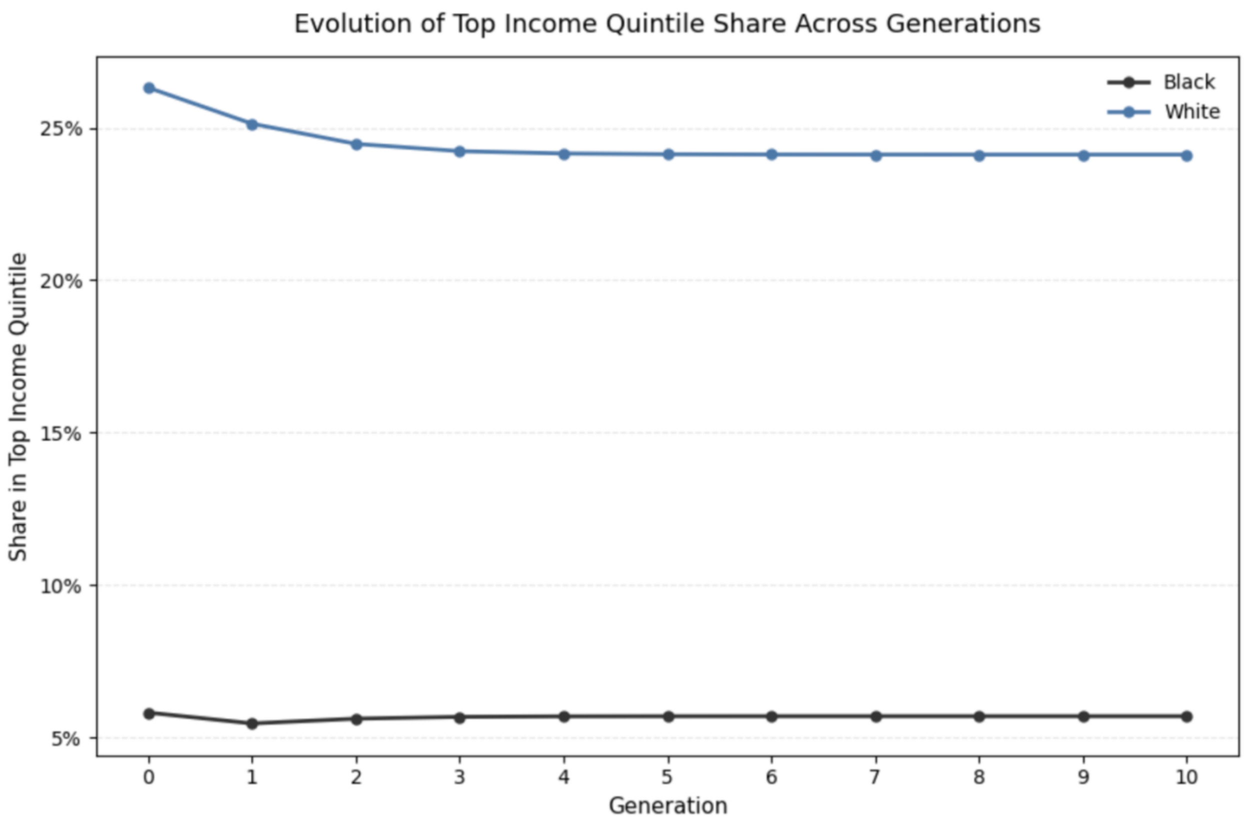
3.4: In a universe where parental income does not influence children's income, the child's income quintile is completely independent of the parent's quintile. Therefore, each column of the transition matrix is identical and equal to the unconditional distribution of children's income. If the unconditional distribution is $(p_1, p_2, p_3, p_4, p_5)$, then the transition matrix has $p_i$in every entry of row i. Parent income plays no role, so the matrix has identical columns.

$$T_{independent} = \begin{pmatrix} p_1 & p_1 & p_1 & p_1 & p_1 \\ p_2 & p_2 & p_2 & p_2 & p_2 \\ p_3 & p_3 & p_3 & p_3 & p_3 \\ p_4 & p_4 & p_4 & p_4 & p_4 \\ p_5 & p_5 & p_5 & p_5 & p_5 \end{pmatrix}$$

In the simplest case, uniform distribution, the matrix would be as follows:

$$T_{independent} = \begin{pmatrix} 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \end{pmatrix}$$

3.5: Plot:



Top Income Quintile Share by Generation

Generation	Black	White
0	0.0581	0.2632
1	0.0545	0.2514
2	0.0561	0.2448
3	0.0567	0.2424
4	0.0569	0.2416
5	0.0569	0.2413
6	0.0569	0.2413
7	0.0569	0.2412
8	0.0569	0.2412
9	0.0569	0.2412
10	0.0569	0.2412

Interpretation of Plot:

The plot shows how the share of children in the top income quintile evolves over ten generations for Black and White families, assuming each generation experiences the same race-specific intergenerational transition matrix as the one estimated in 3.1.

Across generations, the White population maintains a substantially higher share of children in the top quintile. Their top-quintile share begins at 0.2632 and gradually declines toward a long-run value of 0.2412. In contrast, the Black population starts with a much lower top-quintile share, at 0.0569, and remains flat over time. These patterns arise because repeatedly applying the same transition matrix causes each group's income distribution to converge toward the long-run distribution implied by the mobility process itself. Since the White transition matrix features higher upward mobility and a greater likelihood of reaching higher quintiles, its stationary distribution has a larger top-quintile share. Conversely, the Black transition matrix implies a lower steady-state share in the top quintile.

This plot demonstrates that racial disparities in top-quintile representation persist even over many generations. When mobility patterns differ by race, those differences compound over time, and the system converges to distinct long-run outcomes rather than equalizing across groups.